

The sine and cosine rules in three dimensions

Cambridge insert: choosing a triangle inside a solid, drawing it flat, and solving it.

LESSON 14–16

3 × 40 MIN

GEOMETRY 10, §7 (EXTENSION)

IGCSE E6.5 · P1 · 5.5

LESSON CLOCK

10

20'

26'

26'

40'

13'

Recall the two rules	10 min
Choosing the right triangle	20 min
Angle between a line and a plane	26 min
Angle between two planes	26 min
Practice	40 min
Homework	13 min

LEARNING OBJECTIVES

- Identify a triangle inside a solid that contains the required length or angle.
- Redraw that triangle flat, with all known data marked.
- Apply the sine rule, the cosine rule and the area formula in a 3-D context.
- Find the angle between a line and a plane and between two planes.

TEXTBOOK

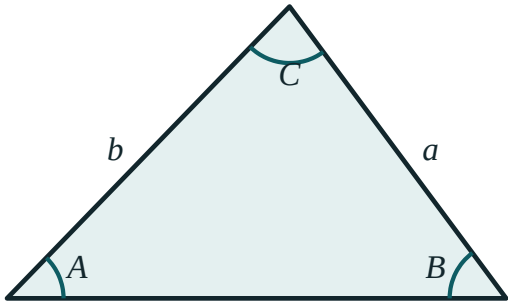
National	pp. 71–78
Cambridge	Core & Extended, pp. 300–312
Workbook	IGCSE Exercise 6.5

01

Explanation

The two rules, recalled

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} \quad a^2 = b^2 + c^2 - 2bc \cdot \cos A$$



$$a / \sin A = b / \sin B = c / \sin C$$
$$a^2 = b^2 + c^2 - 2bc \cdot \cos A$$

The labelling convention: side a opposite angle A.
Every rule assumes it.

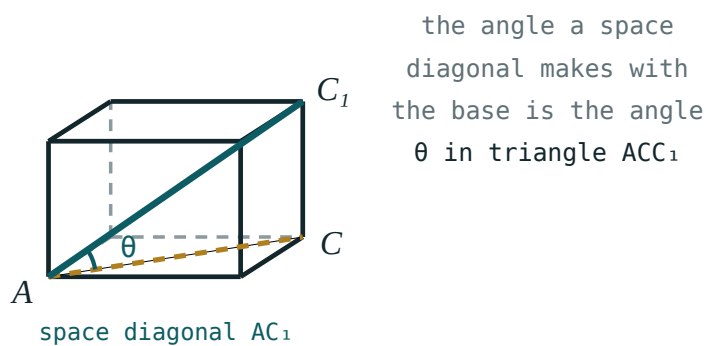
GIVEN	USE
two angles and a side	sine rule
two sides and a non-included angle	sine rule (watch the ambiguous case)
two sides and the included angle	cosine rule
three sides	cosine rule, rearranged for $\cos A$

Area: $S = \frac{1}{2}ab \cdot \sin C$ — two sides and the angle between them.

The method in three dimensions

FOUR STEPS, ALWAYS THE SAME

1 Find a triangle containing what you want. **2** Redraw it flat, large, on its own. **3** Mark every length and angle you already know. **4** Solve it as an ordinary plane triangle.



The box, and beside it the single triangle pulled out and drawn flat.

Almost every 3-D trigonometry question is a plane question hiding inside a solid. The hard part is step 1, and it is made easier by drawing the solid large and marking the right angles first — a vertical edge is perpendicular to **every** line in the base.

The angle between a line and a plane

DEFINITION

Drop a perpendicular from a point of the line to the plane. Join the foot to the point where the line meets the plane. The angle between the line and that **projection** is the angle required — and it is the smallest angle the line makes with any line of the plane.

In a cuboid $a \times b \times c$, the angle θ between the space diagonal and the base:

$$\tan \theta = \frac{c}{\sqrt{a^2 + b^2}}$$

For a $3 \times 4 \times 12$ box: the base diagonal is 5, so $\tan \theta = \frac{12}{5}$ and $\theta \approx 67.4^\circ$.

The angle between two planes

Find the line where the planes meet. From a point on it, draw a line in each plane **perpendicular to that line of intersection**. The angle between those two lines is the dihedral angle.

BOTH LINES MUST BE PERPENDICULAR TO THE EDGE

Any other pair gives the wrong angle. Choosing the edge and then the two perpendiculars is the whole of the technique.

Square pyramid, base 10, height 12: the angle between a lateral face and the base uses the slant height 13 against the apothem 5, giving $\cos \theta = \frac{5}{13}$, so $\theta \approx 67.4^\circ$.

02 Worked examples

EXAMPLE 1 Cuboid $6 \times 8 \times 10$. Find the length of the space diagonal and its angle with the base.

- 1 Base diagonal $\sqrt{36 + 64} = 10$
- 2 Space diagonal $\sqrt{100 + 100} = 10\sqrt{2} \approx 14.14$
- 3 $\tan \theta = \frac{10}{10} = 1$
- 4 $\theta = 45^\circ$

ANSWER

$$10\sqrt{2} \approx 14.1; 45^\circ$$

EXAMPLE 2 A pyramid has a square base of side 8 and lateral edge 10. Find the angle between a lateral edge and the base.

① Half the base diagonal: $\frac{8\sqrt{2}}{2} = 4\sqrt{2} \approx 5.657$

② Right triangle: hypotenuse 10, adjacent 5.657.

③ $\cos \theta = \frac{5.657}{10} = 0.5657$

④ $\theta \approx 55.5^\circ$

ANSWER

$\approx 55.5^\circ$

EXAMPLE 3 In triangle ABC inside a solid, $AB = 7$, $AC = 9$, $\angle BAC = 52^\circ$. Find BC and the area.

① $BC^2 = 49 + 81 - 2 \cdot 7 \cdot 9 \cdot \cos 52^\circ$
Cosine rule.

② $= 130 - 126 \times 0.6157 = 52.42$

③ $BC \approx 7.24$

④ $S = \frac{1}{2} \cdot 7 \cdot 9 \cdot \sin 52^\circ \approx 24.8$

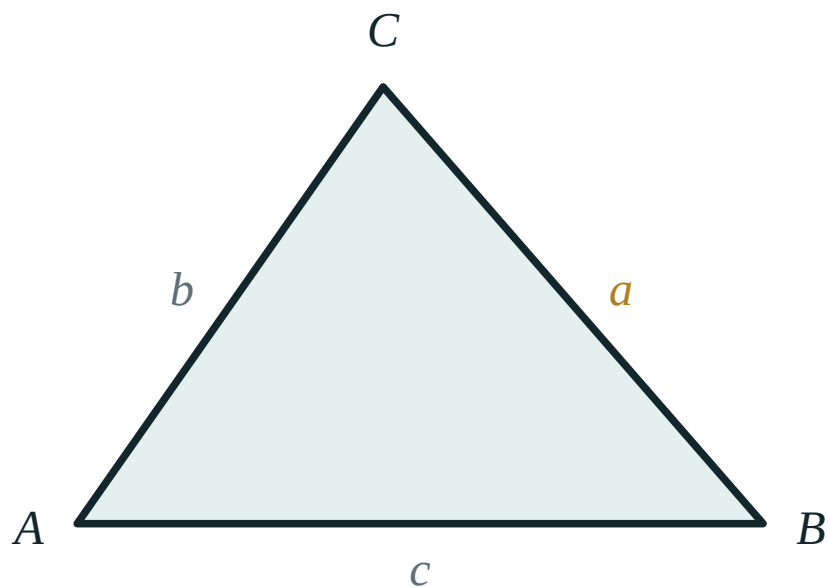
ANSWER

$BC \approx 7.24$, area ≈ 24.8

03 Interactive model

Hold a box and point at the triangle before any calculation begins.

Sine and cosine rules



b 7

c 9

A 55°

a 7.6

B 49° C 76°

area 25.8

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Sine rule	Sinuslar teoremasi	Теорема синусов
Cosine rule	Kosinuslar teoremasi	Теорема косинусов
Included angle	Ichki burchak	Угол между сторонами
Angle between a line and a plane	To'g'ri chiziq va tekislik orasidagi burchak	Угол между прямой и плоскостью
Angle between two planes	Ikki tekislik orasidagi burchak	Угол между плоскостями
Projection	Proyeksiya	Проекция
Foot of the perpendicular	Perpendikulyar asosi	Основание перпендикуляра
Space diagonal	Fazoviy diagonal	Пространственная диагональ
Ambiguous case	Ikki yechimli hol	Неоднозначный случай

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Two sides and the included angle calls for:

the sine rule

the cosine rule

Pythagoras

the area formula

Three sides given, find an angle — use:

the sine rule

the cosine rule rearranged

Pythagoras

a protractor

The angle between a line and a plane is measured to:

any line in the plane

its projection on the plane

the normal

the edge

The space diagonal of an $a \times b \times c$ cuboid is:

$$a + b + c$$

$$\sqrt{a^2 + b^2 + c^2}$$

$$\sqrt{a^2 + b^2}$$

$$abc$$

For a dihedral angle both lines must be perpendicular to:

each other

the line of intersection

the base

the normal

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Space diagonal of a $3 \times 4 \times 12$ box

ANSWER 13

2. Space diagonal of a cube of side 5

ANSWER $5\sqrt{3} \approx 8.66$

3. Base diagonal of a 6×8 rectangle

ANSWER 10

4. Triangle with $a = 5$, $b = 7$, $C = 60^\circ$. Find c

ANSWER $\sqrt{39} \approx 6.24$

5. Area with $a = 6$, $b = 8$, $C = 30^\circ$

ANSWER 12

6. Sine rule: $A = 40^\circ$, $B = 60^\circ$, $a = 10$. Find b

ANSWER ≈ 13.5

7. Angle of a cube's space diagonal with the base

ANSWER $\approx 35.26^\circ$

Medium · the standard question

7 PROBLEMS

1. Cuboid $4 \times 4 \times 7$. Angle of the space diagonal with the base

ANSWER $\approx 51.1^\circ$

2. Square pyramid, base 6, height 4. Angle of a lateral face with the base

ANSWER $\approx 53.1^\circ$

3. Same pyramid: angle of a lateral edge with the base

ANSWER $\approx 43.3^\circ$

4. Triangle $a = 8$, $b = 5$, $c = 7$. Find the largest angle

ANSWER $\approx 81.8^\circ$

5. Triangle $A = 35^\circ$, $a = 9$, $b = 12$. Find B

ANSWER $\approx 49.9^\circ$ or 130.1°

6. Area of a triangle with sides 6, 7 and included angle 75°

ANSWER ≈ 20.3

7. Cube side 4: angle between a face diagonal and a space diagonal

ANSWER $\approx 35.26^\circ$

Hard · stretch and reason

7 PROBLEMS

1. Cuboid $5 \times 12 \times 9$. Angle between the space diagonal and the 12 edge

ANSWER $\approx 44.1^\circ$

2. Square pyramid base 10, lateral edge 13. Find the height

ANSWER $\sqrt{119} \approx 10.9$

3. Same pyramid: dihedral angle between two adjacent faces

ANSWER $\approx 84.8^\circ$

4. A mast stands at a corner of a 20×30 field; from the opposite corner its angle of elevation is 12° . Find its height

ANSWER $\approx 7.66 \text{ m}$

5. Regular tetrahedron edge 6: angle between a face and the base

ANSWER $\approx 70.5^\circ$

6. Regular tetrahedron edge 6: angle between two edges at a vertex

ANSWER 60°

7. Cube: prove the space diagonal makes $\arctan \frac{1}{\sqrt{2}}$ with the base

ANSWER $\tan \theta = \frac{a}{a\sqrt{2}}$

07 Homework

Homework — 6 tasks

Draw the extracted triangle separately for every 3-D question.

1. Cuboid $8 \times 6 \times 10$. Find the space diagonal and its angle with the base.
2. Square pyramid, base 12, height 8. Find the slant height and the angle between a lateral face and the base.
3. Same pyramid: find the angle between a lateral edge and the base.

4. Triangle with $a = 11$, $b = 9$, $C = 48^\circ$. Find c and the area.
5. Triangle with sides 7, 8, 12. Find all three angles.
6. Explain, in three sentences, how to find the angle between two planes.